

Final Project: Priority–Encoded Wavelet Compression for Packet–Lossy Channels

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§1 Abstract

This project implements a traditional Discrete Wavelet Transform image compression system using the Lifting Scheme to address the instability of bitstreams in packet-based radio transmission. Unlike block-based DCT (JPEG) which suffers from catastrophic synchronization loss, DWT allows for multi-resolution sub-band coding.

The core novelty lies in a Priority-Aware Bit-Allocation strategy: critical low-frequency approximation coefficients (LL sub-band) are protected with higher redundancy/priority, while high-frequency detail coefficients (HH, LH, HL) are packetized for "best-effort" delivery. The system is evaluated by simulating non-stationary radio interference and measuring graceful degradation using SSIM metrics.

§2 Background & Related Work

The challenge of transmitting high-quality media over unreliable radio channels is a classic signal processing problem. Traditional standards like JPEG are designed for storage or reliable networks and are notoriously fragile in "packet-erasure" scenarios.

- **The Lifting Scheme:** Proposed by Sweldens (1998), this method revolutionized the Discrete Wavelet Transform (DWT) by breaking convolutions into simple, reversible "Predict" and "Update" steps. It allows for integer-to-integer transforms and in-place computation, making it ideal for hardware-constrained radio systems.
- **Structural Similarity (SSIM):** Developed by Wang et al. (2004), this metric moved quality assessment away from pixel-level error (MSE/PSNR) toward perceptual structural integrity.
- **Priority-Aware Transmission:** Drawing inspiration from Rogers and Cosman (1998), this project utilizes the hierarchical nature of wavelets to apply Unequal Error Protection (UEP), ensuring that "foundation" data survives even when "detail" data is lost.

§3 Methods and Mathematical Framework

Our proposed method replaces the block-based DCT with a Global 2D-DWT to prevent synchronization failure.

§3.1 The Lifting-Based DWT

The transform consists of three primary stages applied row-wise and then column-wise:

1. **Split:** Segregate the signal into even (x_e) and odd (x_o) samples.
2. **Predict:** Use even samples to predict odd ones, keeping only the error (details, d):
$$d[n] = x_o[n] - P(x_e[n])$$
3. **Update:** Use the details to maintain the global average in the even samples (approximation, s):

$$s[n] = x_e[n] + U(d[n])$$

This process is repeated recursively on the **LL quadrant** to create a multi-level pyramid.

§3.2 Priority-Aware Packetization & FEC

We classify the bitstream into two tiers:

- **Tier 1 (High Priority):** Contains the coarsest **LL band** (the image "skeleton"). We protect this with **Forward Error Correction (FEC)** via a **Repetition Code**, where each logical packet is copied multiple times.
- **Tier 2 (Standard Priority):** Contains detail bands (LH, HL, HH). These are sent once without extra protection.

§3.3 Receiver-Side Error Concealment

If a Tier 1 copy survives, the structure is recovered. If a Tier 2 packet is lost, the decoder fills the gap with zeros. This results in "graceful blurring" rather than the "block shifting" seen in JPEG.

§4 Metrics and Results

We evaluated performance using the **Kodak Dataset**¹ and the **SSIM** metric.

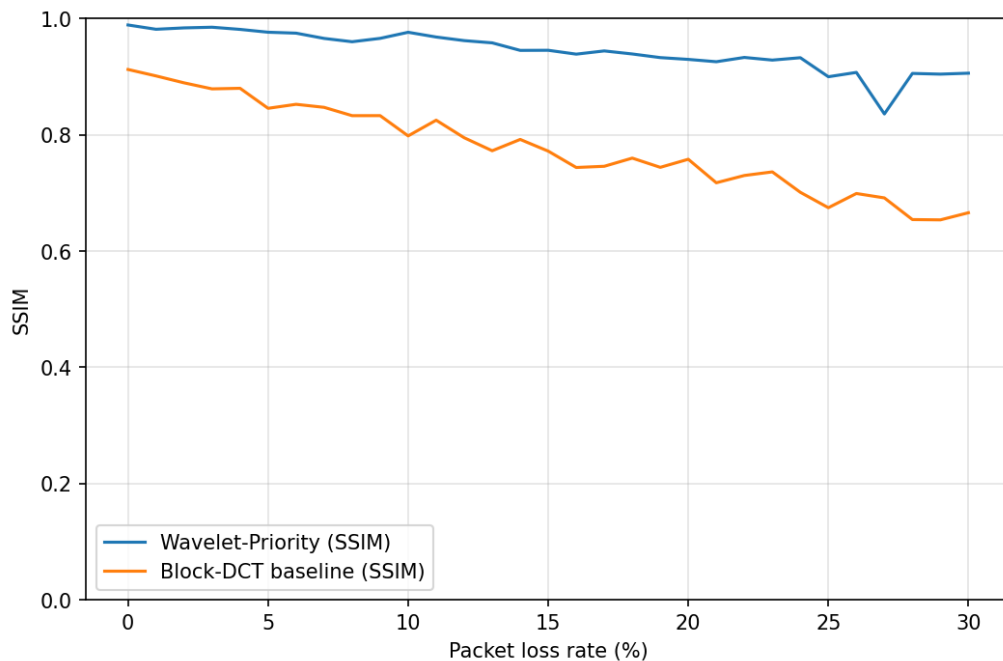


Fig. 1: SSIM v.s Packet loss rate

¹ Kodak Dataset: <https://r0k.us/graphics/kodak/>

kodim03 | loss=0% | W: PSNR=47.12 SSIM=0.989 | DCT: PSNR=34.73 SSIM=0.912



Fig. 2: An example of packet loss rate = 0%

kodim03 | loss=10% | W: PSNR=39.09 SSIM=0.976 | DCT: PSNR=22.09 SSIM=0.798



Fig. 3: An example of packet loss rate = 10%

kodim03 | loss=30% | W: PSNR=31.12 SSIM=0.906 | DCT: PSNR=18.41 SSIM=0.666

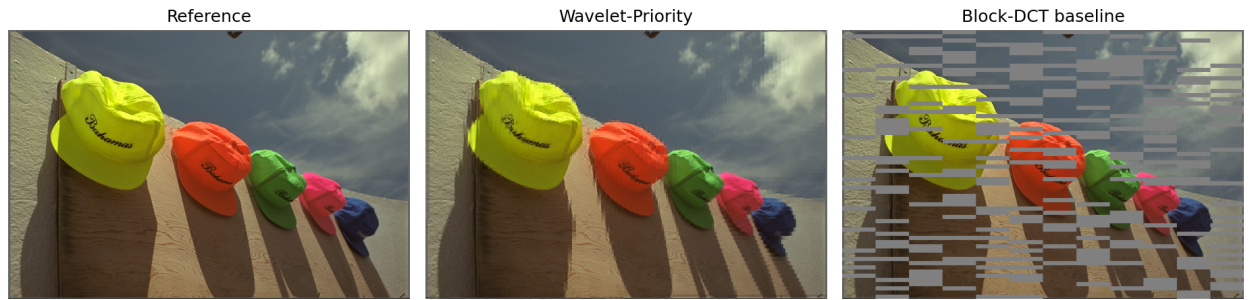


Fig. 4: An example of packet loss rate = 30%

Packet loss rate (%)	Wavelet–Priority (Ours)		Block–DCT baseline	
	PSNR (dB)	SSIM	PSNR (dB)	SSIM
0%	47.12	0.9888	34.73	0.9123
10%	39.09	0.9763	22.09	0.7980
20%	32.34	0.9296	20.78	0.7578
30%	31.12	0.9059	18.41	0.6658

Table 1: PSNR and SSIM metrics for Wavelet–Priority and Block–DCT methods

Our SSIM vs. Packet Loss Rate plot shows that the Wavelet method (Blue) maintains a stable quality curve, whereas the DCT method (Orange) degrades rapidly.

§5 Discussion

The results confirm that structural significance is more important than raw data volume in lossy environments.

- **Result:** The Wavelet method succeeded because its "errors" (missing detail coefficients) are localized in the frequency domain but spread globally in the spatial domain. Conversely, JPEG's DC differential encoding means a single lost packet "poisons" all subsequent blocks.
- **The 0% Loss SSIM Observation:** We noted that at 0% loss, the SSIM was 0.989 (not 1.0). This is due to **Source Coding Loss**—the quantization required to compress the image originally.
- **Expected vs. Ideal:** The repetition code was highly effective but increased the total bit-count. An "ideal" system would use more bandwidth-efficient FEC like Reed–Solomon codes.

§6 Conclusion and Future Work

This project demonstrated that traditional DSP methods like the DWT can provide superior robustness for wireless media transmission when paired with priority-aware logic.

Future Work:

- **Advanced FEC:** Replace repetition codes with **LDPC or Turbo Codes** to achieve the same Tier 1 protection with less bandwidth overhead.
- **Adaptive Quantization:** Dynamically adjust the quantization step based on the detected Signal-to-Noise Ratio (SNR) of the radio channel.
- **Hardware Implementation:** Port the lifting scheme to an FPGA to measure real-time power consumption and throughput on a Jetson Orin or similar edge device.